

Related Work:

Title	Author	Year	Dataset	Application Area	Method	GNN / Graph Type	Metric	Strengths	Limitation	Research Gap	Citations
Hand Gesture Recognition From Surface Electromyography Signals With Graph Convolutional Network and Attention Mechanisms	Zhou et al.	2025	NinaPro DB2: 40 subjects, 12 channels, 49 active gestures; UOW: 10 subjects, 8 sensors, 9 gestures/3 limb positions	sEMG hand-gesture recognition and prosthetic control	CovGCN with covariance-based topology refinement (CovTRM), graph-convolution blocks and temporal modeling	Adaptive covariance-refined weighted GCN; channels are nodes	DB2: 87.5% (Ex-B), 81.9% (C), 93.5% (D); UOW: 84.69%	Exact DB2 evidence; topology and architecture ablations; tests variable limb position	DB2 split is repetition-wise within the same subjects, not held-out-subject; intact users only; offline evaluation	No leakage-free subject-independent DB2 comparison of adaptive covariance graphs against simple anatomical, correlation and fully connected graphs	[1]
Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification	Lee et al.	2025	Five public EMG datasets: NinaPro DB5, BandMyo, UCI, CapgMyo Type A and B	Generalizable and interpretable upper-limb gesture classification	GCN classifier with explicit spatial and spatiotemporal graph representation; sensor-importance and reduced-sensor analysis	KNN-1, KNN-2 and fully connected graphs; optional sensor-wise temporal edges	NinaPro DB5: 92.4% with FC + sensor-wise temporal edges	Systematic topology comparison across five datasets; interpretability; sensor-reduction study	Does not use DB2; the described 8:2 within-subject split limits its value as clean unseen-subject evidence; dense FC+temporal graphs are costly	Effect of explicit spatial-temporal edges on sparse 12-channel DB2 under strict held-out-subject testing remains unknown	[2]
Cross-User Electromyography Pattern Recognition Based on a Novel Spatial-Temporal Graph Convolutional Network	Xu et al.	2024	Private HD-sEMG: 11 subjects, 128 channels, 17 gestures, 8 repetitions	User-independent and transfer-learning-based cross-user myoelectric control	CNN-MSTGCN and TL-CMSTGCN with CNN features, multi-view graphs, spatial-temporal attention and graph/temporal convolution	Multi-view spatial-temporal GCN: distance graph + learned functional graph	68.0% user-independent; 92.3% after target-user transfer learning	True leave-one-subject-out evaluation; ablation; statistical significance tests; low offline test time	HD-sEMG rather than sparse DB2; only 11 subjects/17 gestures; best score requires target-user calibration; no online test	Calibration-free generalization on the complete 40-subject, 49-class, 12-electrode DB2 task is still insufficiently studied	[3]

A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals	Vipulanandan et al.	2024	Private MyoBand: 8 subjects, 8 electrodes, 5 gestures, 20 repetitions, 200 Hz	Real-time prosthetic and augmented-reality gesture recognition	Window-wise graph generation, gesture-event detection and a simple GNN classifier	Fully connected undirected weighted correlation graph; electrodes are nodes	98.19% at 90% overlap; 99.89% at 97.5%; 48 ms end-to-end	Complete real-time-oriented pipeline; reports graph time, prediction latency, ROC and overlap sensitivity	Small private 5-gesture dataset; random 20% graph split and extreme overlap risk near-duplicate/subject leakage; subject-specific threshold	Real-time GNN feasibility has not been shown on a public full-gesture dataset with unseen subjects and leakage-resistant window splitting	[4]
Dynamic Graph Tranformer-Convolutional Network for Multi-Channel EMG Gesture Recognition	Wang et al.	2025	GRABMyo: 43 participants, 3 non-consecutive days, 14 electrodes, 16 gestures, 7 repetitions	Multi-channel EMG hand-gesture recognition	GTCN combining dynamic GCN, four-layer Transformer encoder, positional encoding and attentional feature fusion	Dynamic Pearson-correlation graph + Transformer temporal modeling	97.24% five-fold CV; CNN 94.68%; CNN-LSTM 94.71%	Large multi-day dataset; jointly models changing channel relations and long-range temporal dependencies	Participant/day grouping of folds is unclear; mainly accuracy is reported; no matched static-graph/no-graph ablation; comparatively complex	It remains unclear whether dynamic graphs give reliable unseen-subject gains over static graphs on DB2 and whether gains justify added cost	[5]

Consolidated Research Gap

Recent EMG-GNN studies demonstrate the value of spatial, temporal, covariance-based, functional and dynamic graph structures. However, no reviewed study establishes whether an explicitly constructed electrode graph improves the complete 40-subject, 49-active-class and 12-electrode NinaPro DB2 task under a strictly subject-independent and leakage-free protocol. The literature also lacks a controlled comparison of anatomical, training-only correlation, fully connected and adaptive graphs while holding windowing, node features, subject folds and model capacity constant. Therefore, the proposed work should split subjects before windowing and graph estimation, compare graph and no-graph baselines under matched settings, report macro-F1 and per-class recall, test statistical reliability across grouped subject folds, and evaluate whether any gain justifies added parameters and inference cost.

References

- [1] H. Zhou, H. T. Le, S. Zhang, S. L. Phung, and G. Alici, "Hand Gesture Recognition From Surface Electromyography Signals With Graph Convolutional Network and Attention Mechanisms," *IEEE Sensors Journal*, vol. 25, no. 5, pp. 9081-9092, 2025. doi: 10.1109/JSEN.2025.3527918.
- [2] H. Lee, M. Jiang, J. Yang, Z. Yang, and Q. Zhao, "Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 404-419, 2025. doi: 10.1109/TNSRE.2024.3523943.
- [3] M. Xu, X. Chen, Y. Ruan, and X. Zhang, "Cross-User Electromyography Pattern Recognition Based on a Novel Spatial-Temporal Graph Convolutional Network," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 72-82, 2024. doi: 10.1109/TNSRE.2023.3342050.
- [4] P. Vipulanandan, K. Premaratne, and M. Murthi, "A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals," in *46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1-6, 2024. doi: 10.1109/EMBC53108.2024.10781990.
- [5] P. Wang, T. Xie, and R. H. M. Chan, "Dynamic Graph Transformer-Convolutional Network for Multi-Channel EMG Gesture Recognition," in *47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1-5, 2025. doi: 10.1109/EMBC58623.2025.11253459.